The Journal of Open Source Software

aPhyloGeo: a multi-platform Python package for analyzing phylogenetic trees with climatic parameters

My-Linh Luu[2], **Georges Marceau**[2], **David Beauchemin**[2], **and Nadia Tahiri**[1¶]

1 département d'Informatique, Université de Sherbrooke, 2500 Boulevard de l'Université, Sherbrooke, Québec J1K 2R1, Canada **2** département d'Informatique, Université du Québec à Montréal, 201, avenue du Président-Kennedy, Montréal, Québec H2X 3Y7, Canada ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

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Editor: Frederick Boehm

Reviewers:

- @annazhukova
- @mmore500

Submitted: 06 March 2024
Published: unpublished

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This article and software are linked with research article DOI 10.3847/xxxxx <- update this with the DOI from AAS once you know it., published in the Astrophysical Journal <- The name of the AAS journal..

Abstract

aPhyloGeo is a Python library that explores the relationship between species evolution and environmental pressures, particularly climate. By integrating genetic and climatic data, it helps researchers investigate evolutionary adaptations and identify genetic regions influenced by environmental factors.

The core feature of the software is a comprehensive phylogenetic analysis pipeline with three investigation levels: 1) genetic relationships, 2) climatic impact assessment, and 3) biogeographic correlations. This approach facilitates understanding how species adapt to their environments.

aPhyloGeo employs algorithms using metrics like least squares (Felsenstein, 1997), Euclidean, and Robinson-Foulds (Robinson & Foulds, 1981) distances to ensure statistically sound correlations. Its modular structure offers flexibility, and its open-source nature promotes collaboration.

Available as a PyPI package, *aPhyloGeo* enhances understanding of evolutionary processes and informs conservation efforts, helping prioritize species and populations for preservation in the face of climate change.

Statement of Need

Phylogeography, the study of the interplay between evolution and environment, has long been hindered by the challenges of integrating complex phylogenetic and environmental datasets. aPhyloGeo revolutionizes this field by offering a user-friendly Python package that streamlines phylogeographic analysis. By facilitating the investigation of the intricate relationship between genetic variation and environmental factors, aPhyloGeo is poised to advance our understanding of evolution, biogeography, and conservation efforts. This powerful package makes phylogeographic analysis more accessible and efficient, opening new avenues for research and discovery. The aPhyloGeo package is designed for versatile integration, offering flexibility to be incorporated into various environments. It can be seamlessly embedded within Python scripts for custom workflows, deployed on cloud platforms for scalable analysis, or integrated into web and desktop applications for user-friendly access. This adaptability makes it a valuable tool for diverse research and development needs, from individual projects to large-scale applications.

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Statement of Field

While BEAST (Drummond & Rambaut, 2007) and MrBayes (Ronquist et al., 2012) are primarily used for phylogenetic analyses, they lack integrated tools for analyzing climatic data. Other software like ArcGIS (Scott & Janikas, 2009) focuses on geospatial analysis but may not be tailored for phylogeographic research. aPhyloGeo bridges this gap by providing a unified platform for both analyses, with a streamlined workflow.

aPhyloGeo: A Software Solution

aPhyloGeo is a software package designed for detailed phylogeographic analysis, considering both genetic and climatic data.

Addressing Critical Questions

Given the urgency of climate change, new bioinformatics tools are needed. *aPhyloGeo* will help researchers address critical questions like:

- How has past climate shaped current genetic diversity and distribution?
- How will species distributions evolve under future climates?
- Which populations are most vulnerable to extinction?
- Are there genetic signatures of local adaptation?

Contribution and Impact

aPhyloGeo aims to provide robust tools for analyzing and interpreting phylogeographic data. *aPhyloGeo* will help unravel the complex interplay between genetics, geography, and environment, informing conservation strategies and predicting the impacts of climate change.

Pipeline

Navigating the *aPhyloGeo* workflow (refer to Figure 1) is indispensable to fully harness the potential of this bioinformatics pipeline. The visual representation in Figure 1 outlines the key steps for conducting phylogeographic analysis with optimized effectiveness. For a more detailed understanding, please refer to the comprehensive tutorial provided with the software.

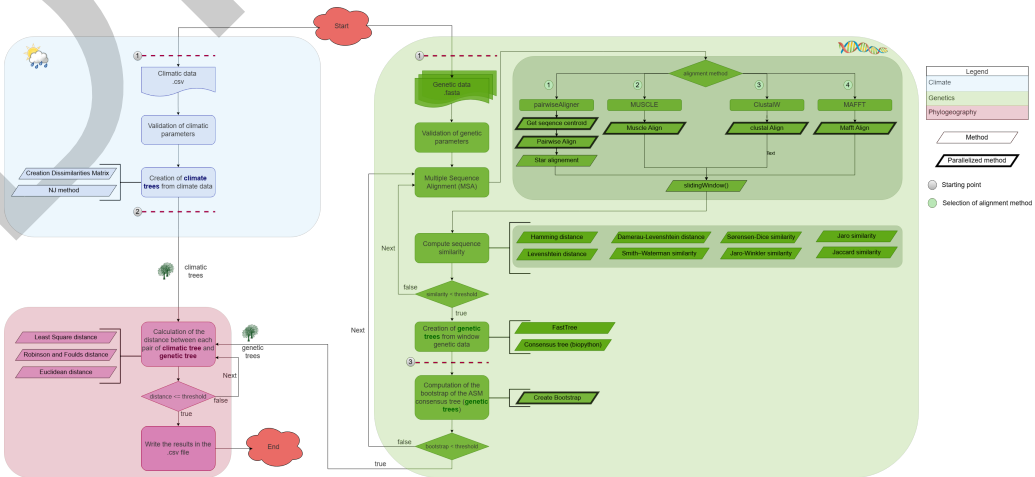

Figure 1: The workflow of the algorithm. The operations within this workflow include several blocks.

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The diagram below illustrates the workflow of the algorithm, consisting of several key blocks, each highlighted with a distinct color (refer to the wiki page).

Multiprocessing

The algorithm supports multiprocessing, allowing the simultaneous analysis of multiple sliding windows (i.e., corresponds to the alignment sub-sequence where the starting point is located after the first position of the alignment, and the ending point is located before the last position of the alignment) within the genetic data. This feature is particularly recommended for large datasets, as it significantly speeds up the analysis by dividing the input sequences into smaller chunks that can be processed in parallel.

Dependencies

This work relies on the following main software packages (refer to pyproject.toml file):

- ete3 (Huerta-Cepas et al., 2016).
- Bio (Cock et al., 2009).

Methods

The methods validate input parameters to ensure data accuracy and prevent errors (refer to the YAML file). ## Tree Comparison

In the comparison of phylogenetic trees, which are constructed based on genetic data, with climatic trees, a crucial step involves applying a phylogeography approach. This includes the utilization of Robinson and Foulds distance for topology evaluation and Least Squares distance for assessing branch length differences.

Editing Multiple Sequence Alignment Methods

Multiple Sequence Alignment (MSA) holds immense significance in bioinformatics as it serves as a foundational step for the comparison and analysis of biological sequences. Here is an in-depth overview of some widely used MSA methods: 1) **Pairwise Alignment** (H. Li, 2018), 2) **MUSCLE** (Edgar, 2004), 3) **CLUSTALW** (Hung & Weng, 2016), and **MAFFT** (Katoh & Standley, 2013).

Similarity Methods

Sequences with notable variability were specifically retained for analysis. The dissimilarity assessment between each sequence pair involved the application of an extensive set of 8 metrics: 1) **Hamming distance** (Labib et al., 2019), 2) **Levenshtein distance** (Yujian & Bo, 2007), 3) **Damerau-Levenshtein distance** (Zhao & Sahni, 2019), 4) **Jaro similarity** (Pradhan et al., 2015), 5) **Jaro-Winkler similarity** (Pradhan et al., 2015), 6) **Smith–Waterman similarity** (Waterman & Smith, 1978), 7) **Jaccard similarity** (Bag et al., 2019), and 8) **Sørensen-Dice similarity** (X. Li et al., 2020).

Conclusion

The *aPhyloGeo* pipeline provides an integrative framework for advanced analysis of genetic and climatic data, promoting simplicity and reproducibility. Future enhancements include clustering techniques and improved alignment methods, alongside novel metrics like Quarted and bipartition distances for better genetic diversity assessment. Emphasizing software development best practices and open-source principles (e.g., iPhyloGeo++), *aPhyloGeo* ensures reliability and sets the stage for ongoing innovation. By prioritizing user empowerment and comprehensive

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analysis, it aims to advance phylogeographic research and deepen our understanding of species-environment interactions.

Acknowledgements

This work was supported by the Natural Sciences and Engineering Research Council of Canada and Fonds de recherche du Québec – Nature et technologies. The computations were performed on resources provided by Compute Canada and Compute Quebec, the national and provincial infrastructure for high-performance computing and data storage. The authors would like to thank the students of the University of Sherbrooke and the Université du Québec à Montréal for their significant contributions to the development of the software. Finally, the authors would like to thank the reviewers and the editor for their valuable comments.

References

Bag, S., Kumar, S. K., & Tiwari, M. K. (2019). An efficient recommendation generation using relevant jaccard similarity. *Information Sciences*, *483*, 53–64. https://doi.org/10.1016/j.ins.2019.01.023

Cock, P. J., Antao, T., Chang, J. T., Chapman, B. A., Cox, C. J., Dalke, A., Friedberg, I., Hamelryck, T., Kauff, F., Wilczynski, B., & others. (2009). Biopython: Freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, *25*(11), 1422. https://doi.org/10.1093/bioinformatics/btp163

Drummond, A. J., & Rambaut, A. (2007). BEAST: Bayesian evolutionary analysis by sampling trees. *BMC Evolutionary Biology*, *7*, 1–8. https://doi.org/10.1186/1471-2148-7-214

Edgar, R. C. (2004). MUSCLE: A multiple sequence alignment method with reduced time and space complexity. *BMC Bioinformatics*, *5*, 1–19. https://doi.org/10.1186/1471-2105-5-113

Felsenstein, J. (1997). An alternating least squares approach to inferring phylogenies from pairwise distances. *Systematic Biology*, *46*(1), 101–111. https://doi.org/10.2307/2413638

Huerta-Cepas, J., Serra, F., & Bork, P. (2016). ETE 3: Reconstruction, analysis, and visualization of phylogenomic data. *Molecular Biology and Evolution*, *33*(6), 1635–1638. https://doi.org/10.1093/molbev/msw046

Hung, J.-H., & Weng, Z. (2016). Sequence alignment and homology search with BLAST and ClustalW. *Cold Spring Harbor Protocols*, *2016*(11), pdb–prot093088. https://doi.org/10.1101/pdb.prot093088

Katoh, K., & Standley, D. M. (2013). MAFFT multiple sequence alignment software version 7: Improvements in performance and usability. *Molecular Biology and Evolution*, *30*(4), 772–780. https://doi.org/10.1093/molbev/mst010

Labib, K., Uznanski, P., & Wolleb-Graf, D. (2019). Hamming distance completeness. *30th Annual Symposium on Combinatorial Pattern Matching (CPM 2019)*, *128*, 14. https://doi.org/10.4230/LIPIcs.CPM.2019.14

Li, H. (2018). Minimap2: Pairwise alignment for nucleotide sequences. *Bioinformatics*, *34*(18), 3094–3100. https://doi.org/10.1093/bioinformatics/bty191

Li, X., Wang, C., Zhang, X., & Sun, W. (2020). Generic sao similarity measure via extended sørensen-dice index. *IEEE Access*, *8*, 66538–66552. https://doi.org/10.1109/ACCESS.2020.2984024

Pradhan, N., Gyanchandani, M., Wadhvani, R., & others. (2015). A review on text similarity technique used in IR and its application. *International Journal of Computer Applications*, *120*(9), 29–34. https://doi.org/10.5120/21257-4109

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Robinson, D. F., & Foulds, L. R. (1981). Comparison of phylogenetic trees. *Mathematical Biosciences*, *53*(1-2), 131–147. https://doi.org/10.1016/0025-5564(81)90043-2

Ronquist, F., Teslenko, M., Van Der Mark, P., Ayres, D. L., Darling, A., Höhna, S., Larget, B., Liu, L., Suchard, M. A., & Huelsenbeck, J. P. (2012). MrBayes 3.2: Efficient bayesian phylogenetic inference and model choice across a large model space. *Systematic Biology*, *61*(3), 539–542. https://doi.org/10.1093/sysbio/sys029

Scott, L. M., & Janikas, M. V. (2009). Spatial statistics in ArcGIS. In *Handbook of applied spatial analysis: Software tools, methods and applications* (pp. 27–41). Springer. https://doi.org/10.1007/978-3-642-03647-7_2

Waterman, M. S., & Smith, T. F. (1978). On the similarity of dendrograms. *Journal of Theoretical Biology*, *73*(4), 789–800. https://doi.org/10.1016/0022-5193(78)90137-6

Yujian, L., & Bo, L. (2007). A normalized levenshtein distance metric. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *29*(6), 1091–1095. https://doi.org/10.1109/TPAMI.2007.1078

Zhao, C., & Sahni, S. (2019). String correction using the damerau-levenshtein distance. *BMC Bioinformatics*, *20*, 1–28. https://doi.org/10.1186/s12859-019-2819-0

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